

Multi-Dimensional Packet Scrambler

Coding for Artificial Intelligence – Problem Set 3

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Description: This program performs four stages:

1. Validate the packet.
2. Scramble the packet using slicing.
3. Insert SYNC-BIT and remove zeros.
4. Check memory integrity using unpacking.

```
1  '''
2  packet_scrambler
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5
6  This program performs four stages:
7
8  1. Validate the packet
9  2. Scramble the packet using slicing
10 3. Insert SYNC-BIT and remove zeros
11 4. Check memory integrity using unpacking
12 '''
13
14 def multidimensional_packet_scrambler(packet):
15
16     # -----
17     # Stage 1: Input Validation and Test Data
18     # -----
19
20     print("\n--- Stage 1: Input Validation and Test Data ---")
21
22     if packet and len(packet) >= 10:
23         print("Validation passed. Preprocessing packet...")
24     else:
25         print("Validation failed: packet is empty or too short.")
26
27     # -----
28     # Stage 2: Middle-Out Swap
29     # -----
30
31     mid = len(packet) // 2
32
33     front_half = packet[:mid]
34     back_half = packet[mid:]
```

```

35
36     scrambled = back_half[::-1] + front_half
37
38     print("\n--- Stage 2: Middle-Out Swap ---")
39     print("Front Half:", front_half)
40     print("Back Half:", back_half)
41     print("Scrambled:", scrambled)
42
43     print("Original packet:", packet)
44     print("Same object:", id(packet) == id(front_half))
45
46     # -----
47     # Stage 3: In-Place Correction
48     # -----
49
50     middle_index = len(scrambled) // 2
51
52     if type(scrambled[middle_index]) is int:
53         scrambled.insert(middle_index + 1, "SYNC-BIT")
54
55     while 0 in scrambled:
56         scrambled.remove(0)
57
58     print("\n--- Stage 3: In-Place Correction ---")
59     print("Final scrambled list:", scrambled)
60
61     # -----
62     # Stage 4: Memory Integrity Check
63     # -----
64
65     first, *middle, last = scrambled
66
67     print("\n--- Stage 4: Memory Integrity Check ---")
68     print("Original packet:", packet)
69     print("Final scrambled:", scrambled)
70     print(
71         f"Header: {first} "
72         f"Footer: {last} "
73         f"Body length: {len(middle)}"
74     )
75
76
77 packet = [5, 12, 0, 8, 21, 34, 7, 19, 0, 3]
78
79 multidimensional_packet_scrambler(packet)

```